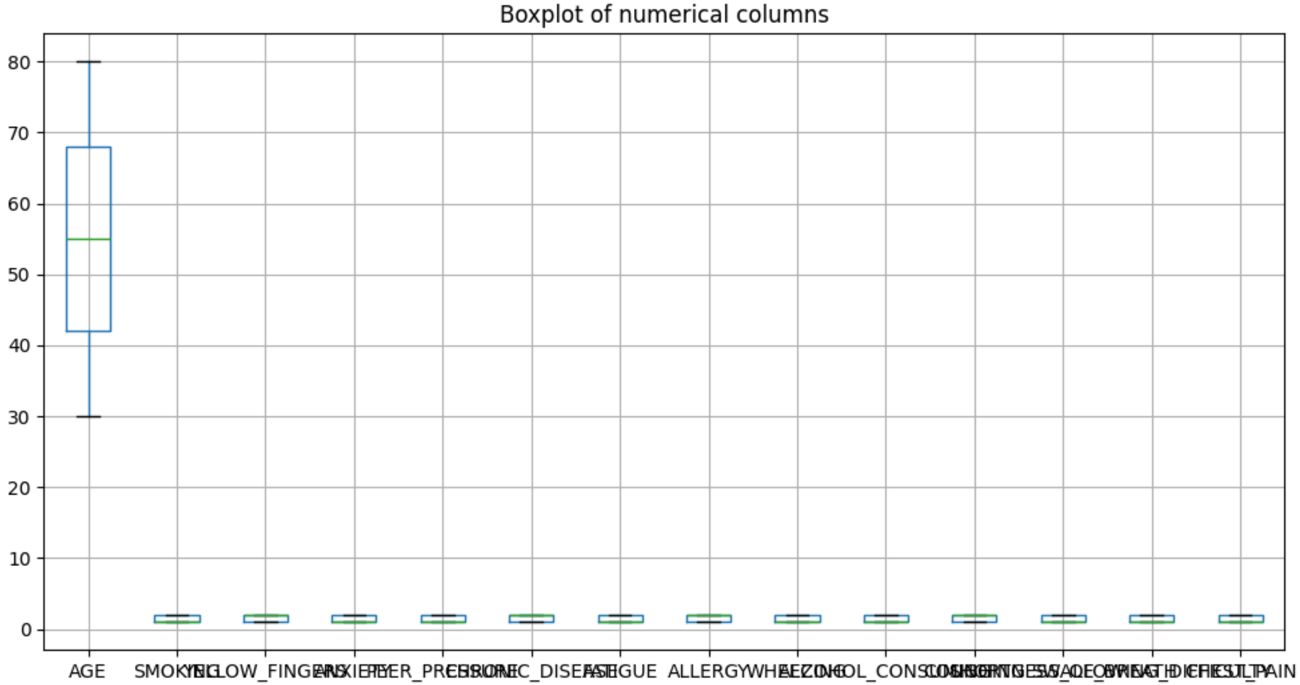
Important: Text generated by AI and may contain incorrect information, always check the content.
To identify outliers, we can use statistical methods and visualizations. Let's start by examining the numerical columns in the DataFrame.

```python
import pandas as pd
import matplotlib.pyplot as plt

# Select numerical columns
numerical_cols = ['AGE', 'SMOKING', 'YELLOW_FINGERS', 'ANXIETY', 'PEER_PRESSURE',
                  'CHRONIC_DISEASE', 'FATIGUE', 'ALLERGY', 'WHEEZING', 'ALCOHOL_CONSUMING',
                  'COUGHING', 'SHORTNESS_OF_BREATH', 'SWALLOWING_DIFFICULTY', 'CHEST_PAIN']

# Calculate summary statistics for numerical columns
print('Summary statistics for numerical columns:')
display(df[numerical_cols].describe())

# Check for outliers using boxplots
print('Boxplots to check for outliers:')
plt.figure(figsize=(12,6))
df[numerical_cols].boxplot()
plt.title('Boxplot of numerical columns')
plt.show()

# Check for outliers using IQR method
Q1 = df[numerical_cols].quantile(0.25)
Q3 = df[numerical_cols].quantile(0.75)
IQR = Q3 - Q1

print('Outliers using IQR method:')
outliers = ((df[numerical_cols] < (Q1 - 1.5 * IQR)) | (df[numerical_cols] > (Q3 + 1.5 * IQR))).sum()
display(outliers)
```

Summary statistics for numerical columns:

	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFF
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.0
mean	55.169000	1.491000	1.514000	1.494000	1.499000	1.509667	1.489667	1.506667	1.497333	1.491333	1.510667	1.488000	1.4
std	14.723746	0.500002	0.499887	0.500047	0.500082	0.499990	0.499977	0.500039	0.500076	0.500008	0.499970	0.499939	0.4
min	30.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
25%	42.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
50%	55.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	1.0
75%	68.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0
max	80.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0

Boxplots to check for outliers:




```
Outliers using IQR method:
```

	0
AGE	0
SMOKING	0
YELLOW_FINGERS	0
ANXIETY	0
PEER_PRESSURE	0
CHRONIC_DISEASE	0
FATIGUE	0
ALLERGY	0
WHEEZING	0
ALCOHOL_CONSUMING	0
COUGHING	0
SHORTNESS_OF_BREATH	0
SWALLOWING_DIFFICULTY	0
CHEST_PAIN	0

dtype: int64

```
This code will help us identify any outliers in the numerical columns of the DataFrame. We will use the `describe()` method to get summary statistics, boxplots to visualize the data, and the Interquartil
```